

Cloud Computing Basics & Core Platform Concepts

Complete Implementation Guide (Manual Process)

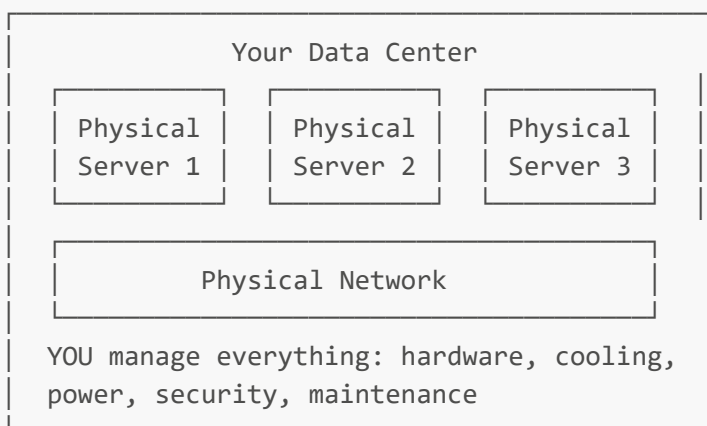
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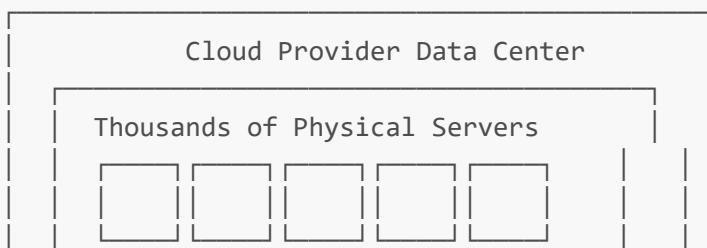
1. Cloud Computing Fundamentals

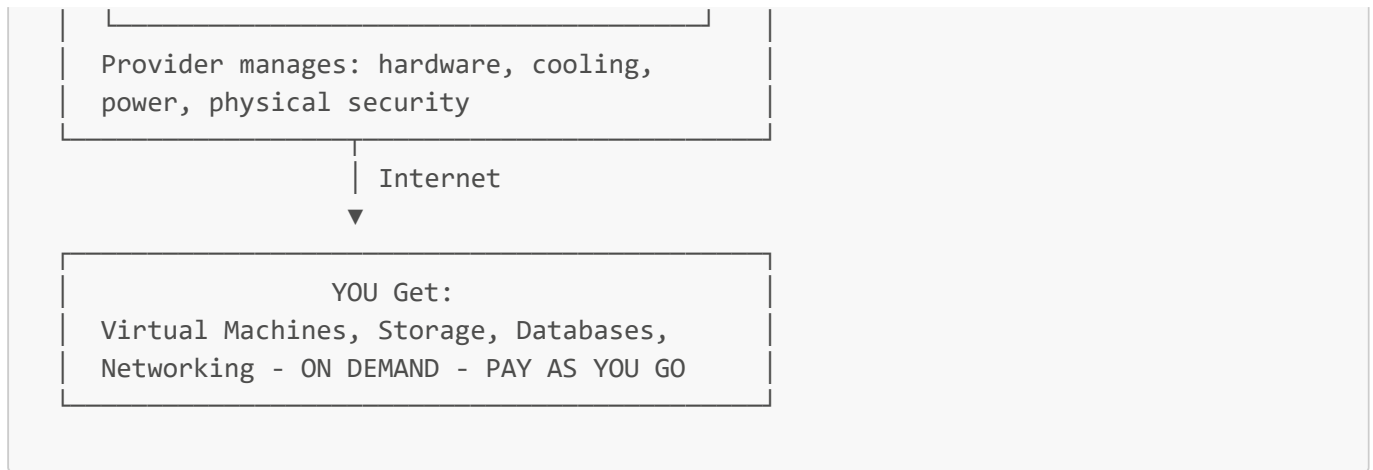
What is Cloud Computing?

Traditional IT Model:



Cloud Model:



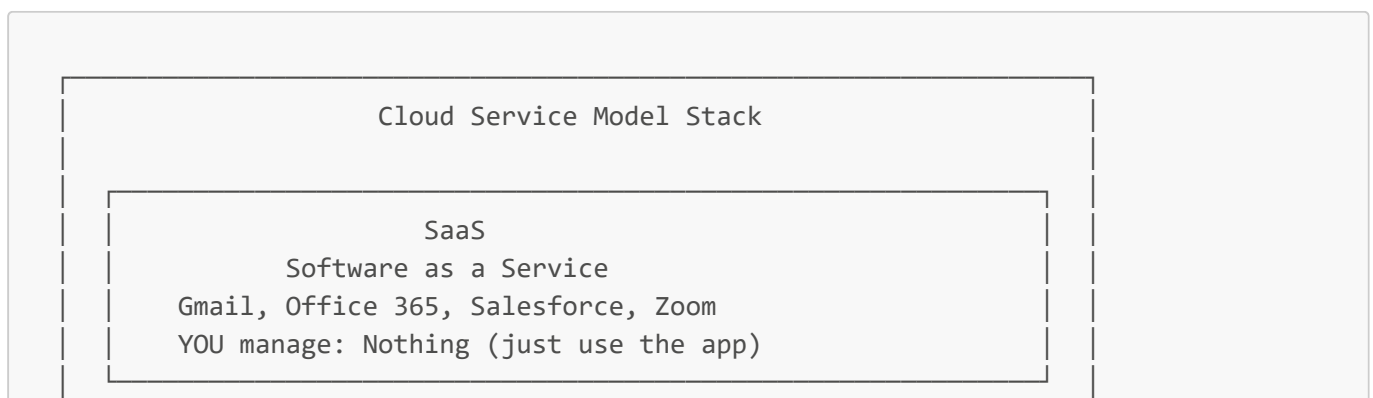


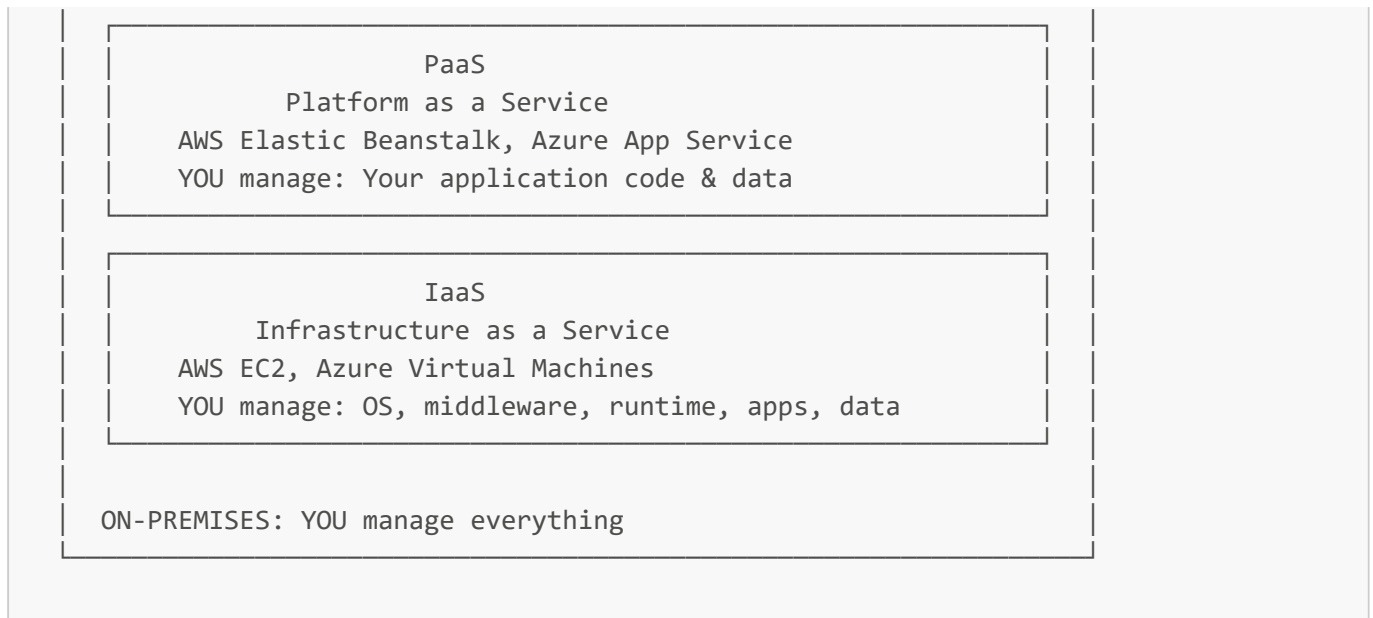
Five Essential Characteristics (NIST Definition)

5 Essential Cloud Characteristics	
Characteristic	What It Means
On-Demand Self-Service	Resources available instantly without human interaction from provider
Broad Network Access	Access from anywhere via internet using standard protocols (HTTP, HTTPS)
Resource Pooling	Provider pools computing resources serving multiple customers (multi-tenancy)
Rapid Elasticity	Scale up or down quickly, sometimes automatically based on demand
Measured Service	Pay only for what you use, like a utility bill (electricity, water)

2. Service Models Deep Dive

The Three Cloud Service Models



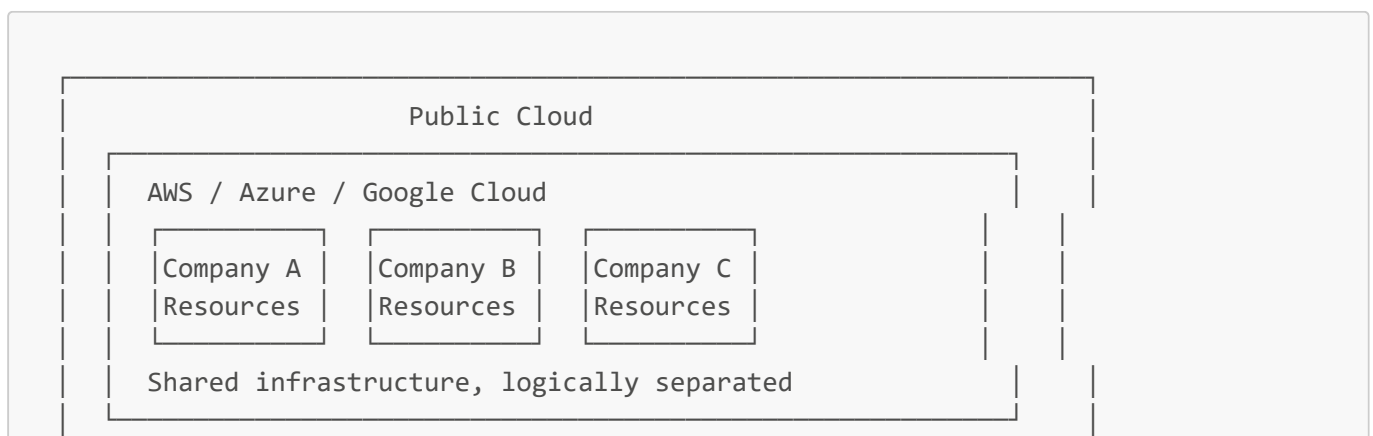


Responsibility Matrix

Component	On-Premises	IaaS	PaaS	SaaS
Applications	YOU	YOU	YOU	Provider
Data	YOU	YOU	YOU	Provider
Runtime	YOU	YOU	Provider	Provider
Middleware	YOU	YOU	Provider	Provider
OS	YOU	YOU	Provider	Provider
Virtualization	YOU	Provider	Provider	Provider
Servers	YOU	Provider	Provider	Provider
Storage	YOU	Provider	Provider	Provider
Networking	YOU	Provider	Provider	Provider
Physical Security	YOU	Provider	Provider	Provider
Facilities	YOU	Provider	Provider	Provider

3. Deployment Models

Four Cloud Deployment Types



- ✓ Cost-effective ✓ No maintenance ✓ Global scale
- ✗ Less control ✗ Shared environment

Private Cloud

Your Organization Only

YOUR dedicated hardware
(on-premises OR hosted by provider)

- ✓ Full control ✓ Better security ✓ Compliance
- ✗ Expensive ✗ You manage hardware

Hybrid Cloud

Private Cloud
(On-Premises)

Sensitive data
Legacy systems

Secure
Link

Public Cloud
(AWS / Azure)

Burst workloads
Dev/Test environments

- ✓ Flexibility ✓ Keep sensitive data local
- ✗ Complex ✗ Requires secure connectivity

Multi-Cloud

AWS

Best-of-breed
services

Azure

Best-of-breed
services

Google Cloud

Best-of-breed
services

- ✓ No vendor lock-in ✓ Best services from each
- ✗ Very complex ✗ Multiple skills needed

4. Core Cloud Benefits

Business Value Framework

Cloud Computing Benefits

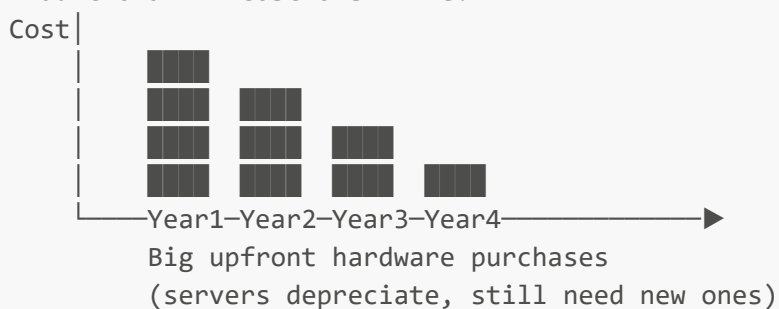


FINANCIAL BENEFITS

- Trade CapEx for OpEx
 - Traditional: Buy servers upfront (\$50,000)
 - Cloud: Pay monthly based on usage (\$500/month)
- Economies of Scale
 - AWS/Azure buy millions of servers = cheaper per unit
 - Savings passed to customers
- Stop Guessing Capacity
 - Traditional: Buy for peak (waste \$ at normal times)
 - Cloud: Scale to exactly what you need
- ⚡ SPEED & AGILITY BENEFITS
 - Deploy in minutes instead of weeks
 - Experiment without risk
 - Go global in minutes
- 🔧 OPERATIONAL BENEFITS
 - Stop spending money on data center operations
 - Focus on YOUR applications, not infrastructure
 - Provider maintains hardware, security patches
- 🌐 RELIABILITY BENEFITS
 - Multiple Availability Zones
 - Built-in redundancy
 - Global content delivery

Cost Model Comparison

Traditional IT Cost Over Time:



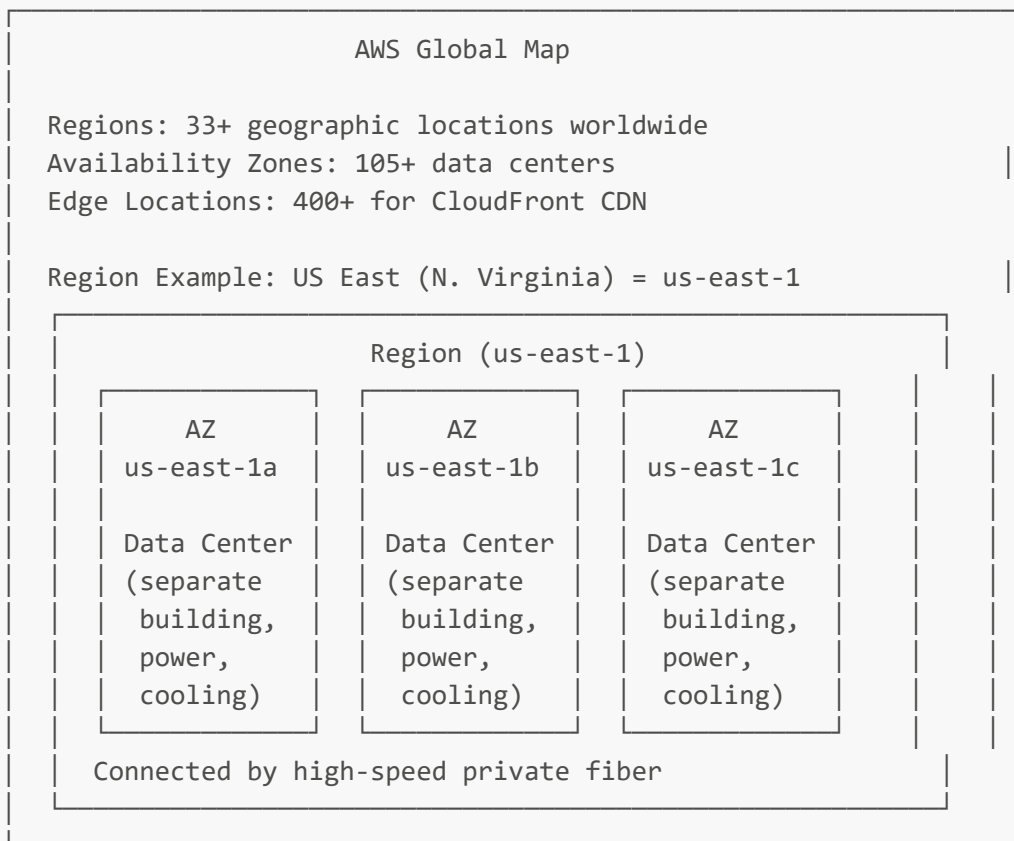
Cloud Cost Over Time:



5. AWS Core Concepts & Manual Setup

AWS Global Infrastructure

AWS Global Infrastructure:



Manual Process: Create Your First AWS Account

STEP 1: Navigate to AWS Account Creation

1. Open browser → go to: <https://aws.amazon.com>
2. Click "Create an AWS Account" (top right corner)

STEP 2: Provide Root User Email

1. Enter your email address (this becomes ROOT USER)
2. Choose an AWS account name
 - ⚠ IMPORTANT: Root user has FULL access to everything
 - ⚠ After setup, create IAM users for daily use

STEP 3: Email Verification

1. Check email for verification code
2. Enter 6-digit code in AWS verification box
3. Click "Verify"

STEP 4: Root User Password

1. Create strong password (min 8 chars)

- ✓ Must include: uppercase, lowercase, number, symbol

2. Confirm password
3. Click "Continue"

STEP 5: Contact Information

1. Select account type:
 - ☐ Business ☐ Personal
2. Fill in:
 - Full name
 - Phone number
 - Country/Region
 - Address, City, State, Postal code
3. Check "AWS Customer Agreement" box
4. Click "Continue"

STEP 6: Payment Information

1. Enter credit/debit card details
 -  Required but won't be charged for free tier usage
 -  AWS free tier gives 12 months free on many services
2. Billing address: same as above or new address
3. Click "Verify and Continue"

STEP 7: Identity Verification

1. Choose verification method:
 - ☐ Text message (SMS)
 - ☐ Voice call
2. Enter phone number
3. Enter CAPTCHA characters shown
4. Click "Send SMS" or "Call Me"
5. Enter received verification code
6. Click "Continue"

STEP 8: Select Support Plan

1. Choose plan:

- | |
|--|
| <ul style="list-style-type: none">● Basic Support (FREE) ← Select this<ul style="list-style-type: none">- Account/billing questions- Service limit increases- AWS documentation access |
| <ul style="list-style-type: none">○ Developer (\$29/month)<ul style="list-style-type: none">+ Business hours email support |
| <ul style="list-style-type: none">○ Business (\$100/month minimum)<ul style="list-style-type: none">+ 24/7 phone, email, chat support |

2. Click "Complete Sign Up"

STEP 9: Account Activation

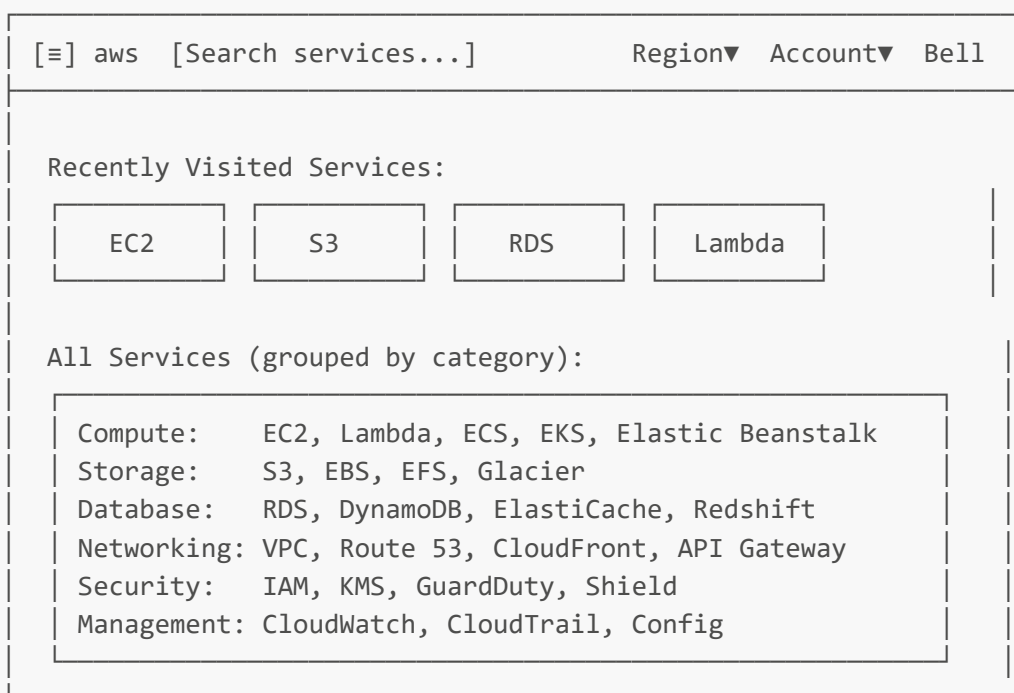
1. Wait 24-48 hours for activation email
2. Usually activates within minutes to a few hours
3. You'll receive email: "Your AWS Account is Ready"

STEP 10: First Login & Console Navigation

1. Go to: <https://console.aws.amazon.com>
2. Sign in as Root user
3. Enter root email → password
4. You're now in AWS Management Console

Understanding AWS Console Layout

AWS Management Console Layout:



Manual Process: AWS IAM Setup (Security Best Practice)

WHY IAM USERS? Never use root account for daily work!
Root account = master key to your entire AWS account
IAM users = limited keys for specific tasks

STEP 1: Navigate to IAM

1. In AWS Console search bar, type: IAM
2. Click "IAM" in results
3. You're now in Identity and Access Management

STEP 2: Enable MFA for Root Account First

1. In IAM Dashboard, look for "Security recommendations"

2. Click "Add MFA for root user"
3. Click "Activate MFA"
4. Choose MFA device type:
 - ☐ Authenticator app (recommended - Google Authenticator)
 - ☐ Security key
 - ☐ Hardware TOTP token
5. If Authenticator app:
 - Open Google Authenticator on phone
 - Tap "+" → "Scan QR code"
 - Scan QR code shown on screen
 - Enter two consecutive 6-digit codes
 - Click "Add MFA"
6. ☒ Root MFA is now enabled

STEP 3: Create Admin IAM User

1. In left sidebar, click "Users"
2. Click "Create user" button
3. Enter user details:
 - User name: admin-yourname (example: admin-john)
 - ☒ Check "Provide user access to AWS Management Console"
 - Select "I want to create an IAM user"
 - Console password: Custom password or Auto-generated
 - ☒ "User must create new password at next sign-in" (optional)
4. Click "Next"

STEP 4: Set Permissions for Admin User

1. Select "Attach policies directly"
2. Search for: AdministratorAccess
3. ☒ Check the box next to "AdministratorAccess"
4. Click "Next"

STEP 5: Add Tags (Optional but Recommended)

1. Add tags to organize resources:

Key: Department	Value: IT
Key: Environment	Value: Production
Key: Owner	Value: YourName
2. Click "Create user"

STEP 6: Save Access Information

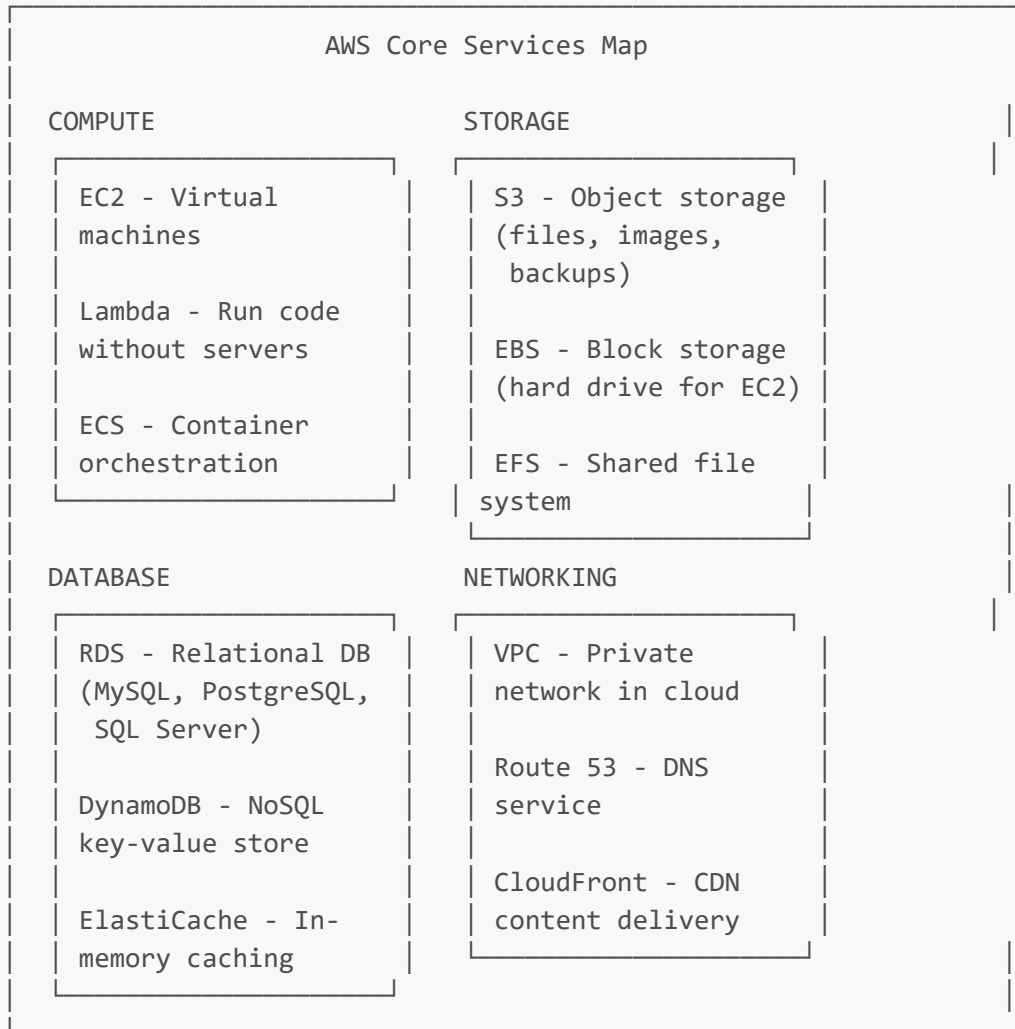
1. Note your:
 - Account ID (12-digit number shown in top right)
 - IAM user sign-in URL:
`https://[account-id].signin.aws.amazon.com/console`
2. Click "Return to users list"

STEP 7: Sign Out and Sign Back In as IAM User

1. Click Account menu → Sign Out
2. Go to IAM user sign-in URL

3. Enter: Account ID, Username, Password
4. ☒ Now you're working as IAM user (not root)

AWS Key Services Overview



Manual Process: Create Your First EC2 Instance

What is EC2?

EC2 = Elastic Compute Cloud

= Virtual computer (server) in the cloud

= You choose: CPU, RAM, storage, OS

STEP 1: Navigate to EC2

1. AWS Console → Search "EC2"
2. Click "EC2" → You're in EC2 Dashboard
3. Click "Launch Instance" (orange button)

STEP 2: Name Your Instance

1. Under "Name and tags":
 - Name: my-first-server
 - This is just a label for organization

STEP 3: Choose an AMI (Operating System)

AMI = Amazon Machine Image = pre-configured OS template

1. Under "Application and OS Images (Amazon Machine Image)":
2. "Quick Start" tab shows common options:

- | |
|--|
| • Amazon Linux 2023 (Free tier eligible) ← Best AWS-optimized, lightweight, secure |
| ○ Ubuntu 22.04 LTS (Free tier eligible)
Popular Linux distribution |
| ○ Windows Server 2022 (Additional cost)
For Windows-based applications |

3. Select: Amazon Linux 2023 AMI
4. Architecture: 64-bit (x86)

STEP 4: Choose Instance Type

Instance Type = Size of your virtual computer

1. Click "Compare instance types" to understand options
2. Instance naming pattern:
t3.micro
 - Size (nano, micro, small, medium, large, xlarge)
 - Generation (newer = better)
 - Family (t=general, c=compute, r=memory, g=GPU)

3. Free tier: Select t2.micro or t3.micro

t2.micro: 1 vCPU, 1 GB RAM - FREE TIER
t2.small: 1 vCPU, 2 GB RAM
t2.medium: 2 vCPU, 4 GB RAM
t3.micro: 2 vCPU, 1 GB RAM - FREE TIER
m5.large: 2 vCPU, 8 GB RAM

STEP 5: Create Key Pair (for SSH access)

Key pair = digital lock/key system to securely access server

1. Under "Key pair (login)":
2. Click "Create new key pair"
3. In popup window:
 - Key pair name: my-server-key
 - Key pair type: • RSA
 - Private key file format: • .pem (for Mac/Linux)
 - .ppk (for Windows PuTTY)

4. Click "Create key pair"
5.  File downloads automatically to your computer
6.  **SAVE THIS FILE!** You cannot download it again!
Store in safe location: ~/.ssh/my-server-key.pem

STEP 6: Configure Network Settings

1. Under "Network settings":
2. Click "Edit" to see all options
3. VPC: Keep default VPC
4. Subnet: Keep default (any Availability Zone)
5. Auto-assign public IP: Enable ← So you can access it
6. Firewall (Security Groups):
 - Select "Create security group"
 - Security group name: my-server-sg
 - Description: Security group for my first server
 - Inbound rules:

Rule 1 (already added): Type: SSH Port: 22 Source: My IP Purpose: Remote access from your computer
--

STEP 7: Configure Storage

1. Under "Configure storage":
2. Default: 8 GiB gp3 (free tier allows up to 30 GiB)
3. Change to 20 GiB for more space
4. Volume type options:
 - gp3: General Purpose SSD (recommended)
 - io1/io2: High performance SSD (more expensive)
 - st1: Throughput optimized HDD (big data)

STEP 8: Review and Launch

1. Review "Summary" panel on right:

Number of instances: 1
AMI: Amazon Linux 2023
Instance type: t2.micro
Key pair: my-server-key
Estimated cost: \$0 (free tier)

2. Click "Launch instance" (orange button)

STEP 9: Verify Instance is Running

1. Click "View all instances"
2. See your instance in the list
3. Wait for "Instance State" = running (green dot)
4. Wait for "Status Check" = 2/2 checks passed
(takes 2-5 minutes)
5. Note your "Public IPv4 address" (example: 54.123.45.67)

Manual Process: Create S3 Bucket

What is S3?

S3 = Simple Storage Service

= Store any file: images, videos, backups, code

= Virtually unlimited storage

= 99.999999999% (11 9's) durability

STEP 1: Navigate to S3

1. AWS Console → Search "S3"
2. Click "S3"
3. Click "Create bucket"

STEP 2: Configure Bucket

1. General configuration:
 - AWS Region: US East (N. Virginia) us-east-1
 - Bucket name: myapp-storage-2024-yourinitials
 - ⚠ Must be globally unique across ALL AWS accounts
 - ⚠ Lowercase letters, numbers, hyphens only
 - ⚠ 3-63 characters long
2. Object Ownership:
 - ACLs disabled (recommended)
 - Bucket owner has full control
 - Simpler security management
3. Block Public Access settings:
 - ☒ Block all public access (recommended for most cases)
 - ⚠ Only UNCHECK if hosting a public website
4. Bucket Versioning:
 - Disable (for learning)
 - Enable (for production - keeps all file versions)
5. Default encryption:
 - Encryption type: SSE-S3 (AWS managed keys)
 - Automatically encrypts all stored objects
6. Click "Create bucket"

STEP 3: Upload Your First File

1. Click on your bucket name
 2. Click "Upload"
 3. Click "Add files"
 4. Select a file from your computer
 5. Under "Properties":
 - Storage class: Standard (for frequently accessed data)
-

Storage Classes and Use Cases:

- Standard: Frequent access (\$0.023/GB/month)
- IA (Infrequent Access): Occasional access
- Glacier: Long-term archive (very cheap)
- Intelligent-Tiering: Auto-moves between tiers based on access patterns

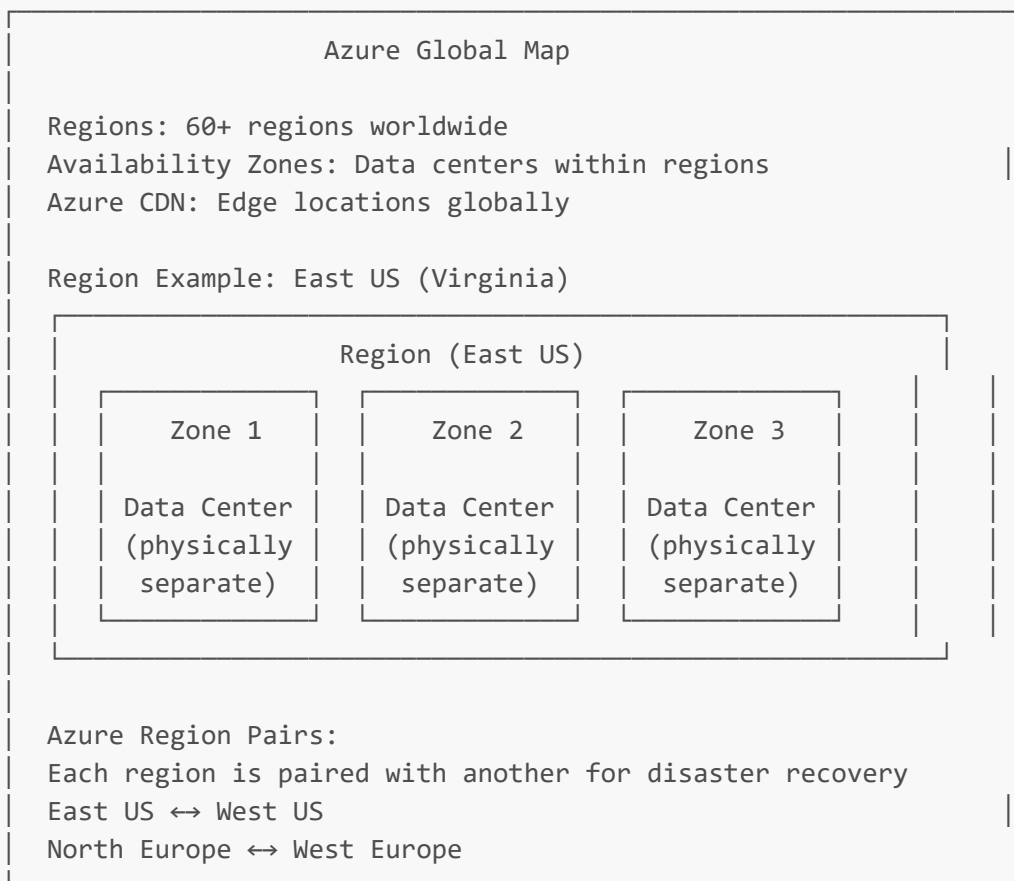
6. Click "Upload"

7. Wait for "Upload succeeded" message

6. Azure Core Concepts & Manual Setup

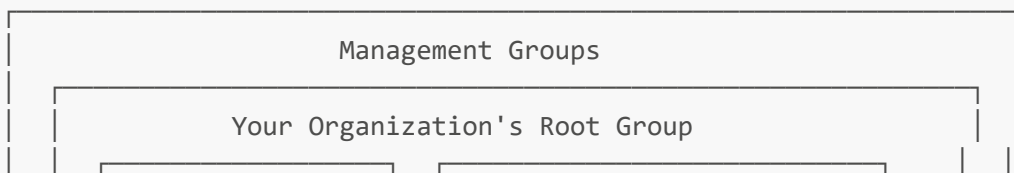
Azure Global Infrastructure

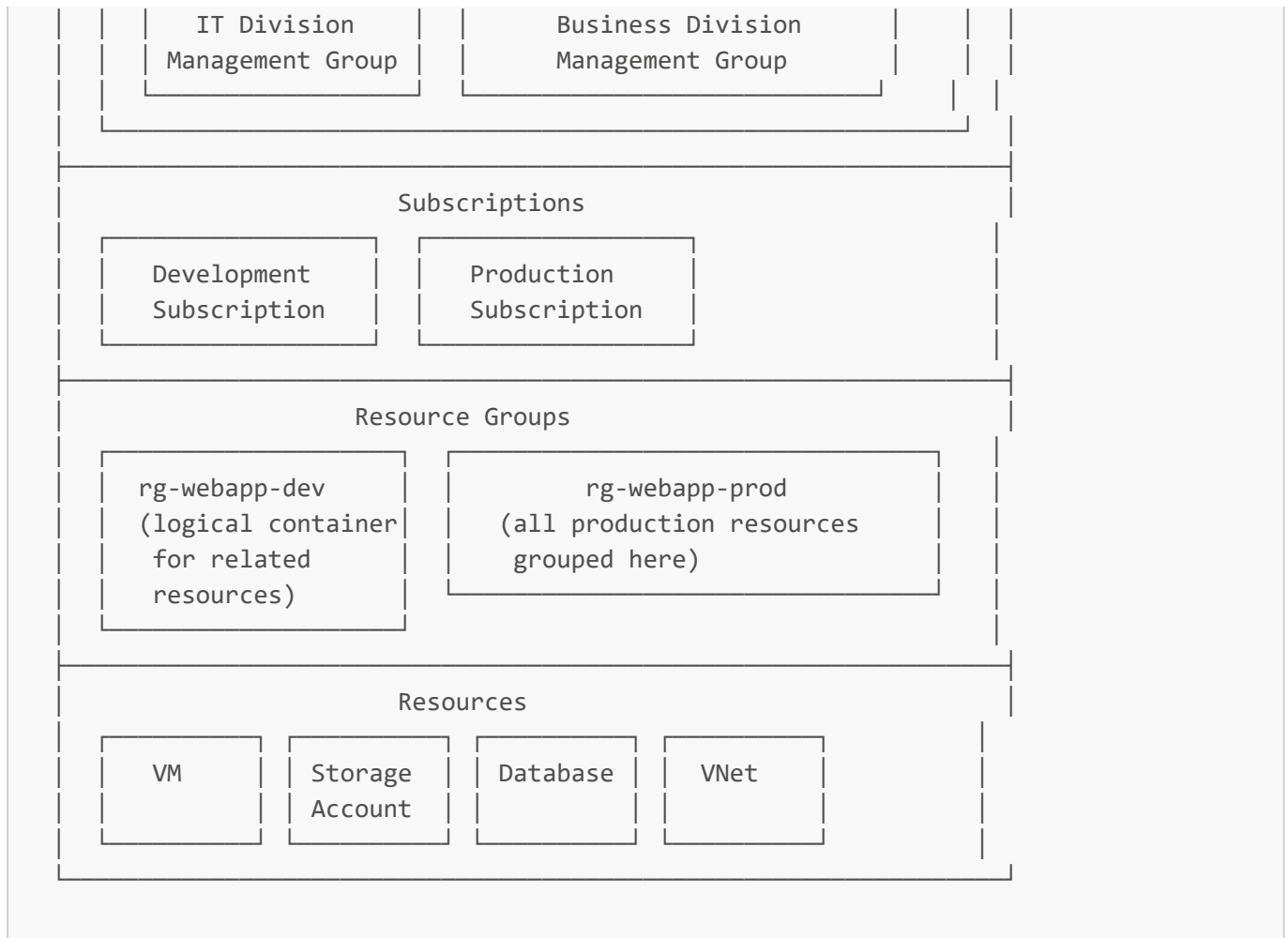
Azure Global Infrastructure:



Azure Hierarchy (CRITICAL CONCEPT)

Azure Resource Organization Hierarchy:





Manual Process: Create Azure Free Account

STEP 1: Navigate to Azure Free Account

1. Open browser → go to: <https://azure.microsoft.com/free>
2. Click "Start free" button

STEP 2: Microsoft Account Sign-In

Option A - Have Microsoft/Outlook account:

1. Click "Sign in"
2. Enter existing Microsoft account email
3. Enter password

Option B - Create new Microsoft account:

1. Click "Create one"
2. Enter new email or use existing email
3. Create password for Microsoft account
4. Verify with code sent to email/phone

STEP 3: Azure Sign-Up Form

1. Your profile tab:
 - Country/Region: Select your country
 - First name, Last name

- Email address
- Phone: Enter mobile number
- Click "Text me" or "Call me" for verification code
- Enter verification code
- Company: Optional

2. Identity verification by phone:

- Country code
- Phone number
- Click "Text me"
- Enter code received

3. Identity verification by card:

- Card number
- Expiration date, CVV
- Name on card, Billing address
-  \$1 temporary hold (refunded in 3-5 days)
-  Won't be charged unless you upgrade to paid

4. Technical support plan:

- No technical support (Free) ← Select this

5. Agreement:

- ☒ I agree to the subscription agreement...
- Click "Sign up"

STEP 4: Welcome to Azure!

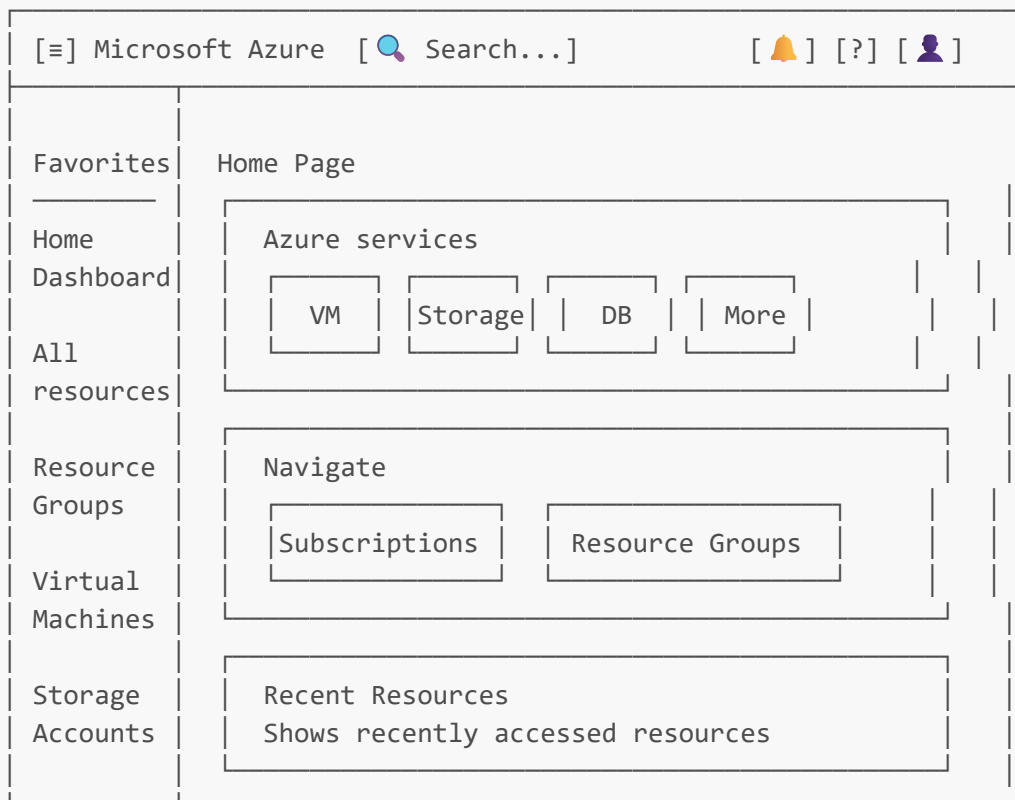
1. Click "Go to the portal" or go to:
<https://portal.azure.com>
2. You're now in the Azure Portal

STEP 5: Azure Free Account Benefits

Azure Free Tier Includes:
\$200 credit for first 30 days 12 months free popular services: - 750 hours Linux/Windows VMs (B1s) - 5 GB Azure Blob Storage - 250 GB SQL Database - 5 GB Azure Cosmos DB Always free services: - 1 million Azure Functions requests/month - 10 web apps (Azure App Service free tier) - Azure DevOps (5 users)

Understanding Azure Portal Layout

Azure Portal Layout:



Manual Process: Create Resource Group in Azure

Resource Group = Logical container for Azure resources
 = Organize related resources together
 = Apply policies, permissions, tags to group

STEP 1: Navigate to Resource Groups

1. Azure Portal (<https://portal.azure.com>)
2. Search bar at top: type "Resource groups"
3. Click "Resource groups" in results
4. Click "+ Create" button

STEP 2: Configure Resource Group

1. Basics tab:

- Subscription: Azure subscription 1 (or your subscription)
- Resource group name: rg-myapp-dev
Naming convention: rg-[project]-[environment]
- Region: (US) East US
 - i** Resource group region = where metadata is stored
 - i** Resources inside can be in different regions

2. Tags tab (recommended for organization):

Name	Value
------	-------

Environment	Development
Project	MyApp
Owner	YourName
CostCenter	IT-001

3. Review + Create tab:

- Review settings summary
- Click "Create" button

STEP 3: Verify Creation

1. Notification bell (🔔) shows "Resource group created"
2. Click "Go to resource group"
3. You'll see empty resource group ready for resources

Manual Process: Create Azure Virtual Machine

STEP 1: Navigate to Virtual Machines

1. Azure Portal → Search "Virtual machines"
2. Click "+ Create" → "Azure virtual machine"

STEP 2: Basics Configuration

1. Project Details:

- Subscription: Your subscription
- Resource group: rg-myapp-dev (select existing)
OR click "Create new" → enter name

2. Instance Details:

- Virtual machine name: vm-myapp-web-01
Naming convention: vm-[project]-[role]-[number]
- Region: (US) East US
- Availability options:
 - No infrastructure redundancy (for learning)
 - Availability Zone (for production)
- Security type: Standard
- Image: Ubuntu Server 22.04 LTS - x64 Gen2
(Or Windows Server 2022 if needed)
- Size: Click "See all sizes"

Recommended free tier options:

B1s: 1 vCPU, 1 GB RAM - FREE TIER

B2s: 2 vCPU, 4 GB RAM

D2s_v3: 2 vCPU, 8 GB RAM (general purpose)

Select: Standard_B1s (free tier)

Click "Select"

STEP 3: Administrator Account

For Linux VM:

- Authentication type: • SSH public key (recommended)
- Username: azureuser
- SSH public key source:
 - Generate new key pair (easiest for beginners)
- Key pair name: vm-myapp-key

⚠ OR use Password:

- Authentication type: • Password
- Username: azureuser
- Password: [Strong password - min 12 chars]
- Confirm password

STEP 4: Inbound Port Rules

1. Public inbound ports: • Allow selected ports
 2. Select inbound ports: SSH (22)
 - ✓ SSH (22) ← Required to connect to Linux VM
 - ✓ HTTP (80) ← For web server access
- ⚠ Only for learning - use Network Security Groups in production

STEP 5: Disks Tab

1. OS disk type:
 - Standard SSD (Locally-redundant storage) ← Free tier
 - Premium SSD (better performance)
 - Standard HDD (cheapest, slowest)
2. Leave other settings as default

STEP 6: Networking Tab

1. Virtual network:
 - (new) rg-myapp-dev-vnet (auto-created)
2. Subnet:
 - (new) default (10.0.0.0/24)
3. Public IP:
 - (new) vm-myapp-web-01-ip
4. NIC network security group: Basic
5. Public inbound ports: (same as previous tab)

STEP 7: Management Tab

1. Microsoft Defender for Cloud:
 - Basic (free) ← for learning
2. Auto-shutdown:
 - Enable → Set time: 11:00 PM (save costs!)
 - Time zone: Your timezone

STEP 8: Review + Create

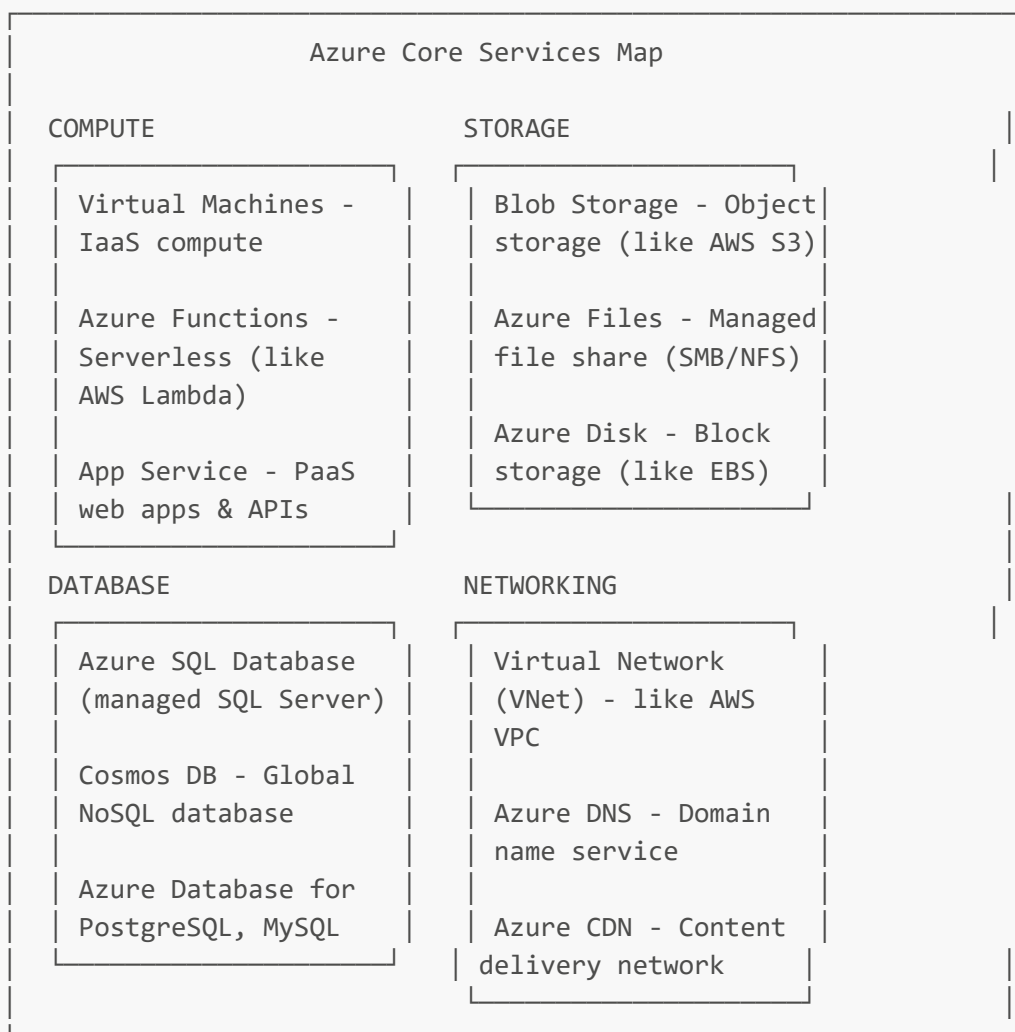
1. Review validation passed (green checkmark)
2. Review estimated cost at bottom

3. Click "Create"
4. If generating SSH key pair:
 - Popup appears: "Generate new key pair"
 - Click "Download private key and create resource"
 - ⬇ Saves .pem file to your computer
 - ⚠ Save this file securely!

STEP 5: Monitor Deployment

1. Deployment page shows progress:
 - Creating virtual machine... (2-5 minutes)
 - Creating public IP address...
 - Creating network interface...
 - Creating virtual network...
2. When complete: "Your deployment is complete" ✓
3. Click "Go to resource"

Azure Key Services Overview



Manual Process: Create Azure Storage Account & Blob Container

STEP 1: Navigate to Storage Accounts

1. Azure Portal → Search "Storage accounts"
2. Click "+ Create"

STEP 2: Basic Configuration

1. Project details:
 - Subscription: Your subscription
 - Resource group: rg-myapp-dev
2. Instance details:
 - Storage account name: mystorageaccount2024
 - ⚠ Must be globally unique (lowercase, numbers only)
 - ⚠ 3-24 characters
 - Region: East US
 - Performance:
 - Standard (HDD-based, most use cases)
 - Premium (SSD-based, high performance)
 - Redundancy:

LRS - Locally Redundant Storage 3 copies in ONE data center (cheapest) Protects against: disk failure
ZRS - Zone Redundant Storage 3 copies across 3 Availability Zones Protects against: data center failure
GRS - Geo-Redundant Storage 6 copies: 3 in primary, 3 in secondary region Protects against: regional disaster
GZRS - Geo-Zone Redundant Storage Most protection (most expensive)

Select: LRS for learning

STEP 3: Advanced Tab (Optional)

1. Require secure transfer (HTTPS): • Enabled ☒
2. Enable blob public access: ○ Disabled (secure default)
3. Minimum TLS version: TLS 1.2 ☒

STEP 4: Review + Create

1. Click "Review"
2. Review settings
3. Click "Create"
4. Wait for deployment (30-60 seconds)
5. Click "Go to resource"

STEP 5: Create Blob Container

1. In Storage Account → Left sidebar: "Containers"
2. Click "+ Container"
3. In popup:
 - Name: images (lowercase, no spaces)
 - Public access level:
 - Private (no anonymous access) ← Recommended
 - Blob (anonymous read for blobs)
 - Container (anonymous read for container)
4. Click "Create"

STEP 6: Upload a Blob (File)

1. Click on "images" container
2. Click "Upload"
3. In upload panel:
 - Click folder icon → select file from computer
 - Advanced options:
 - Authentication type: Account key
 - Blob type: Block blob (for files)
 - Block size: 8 MiB (default)
4. Click "Upload"
5. File appears in container list 

7. AWS vs Azure Architecture Comparison

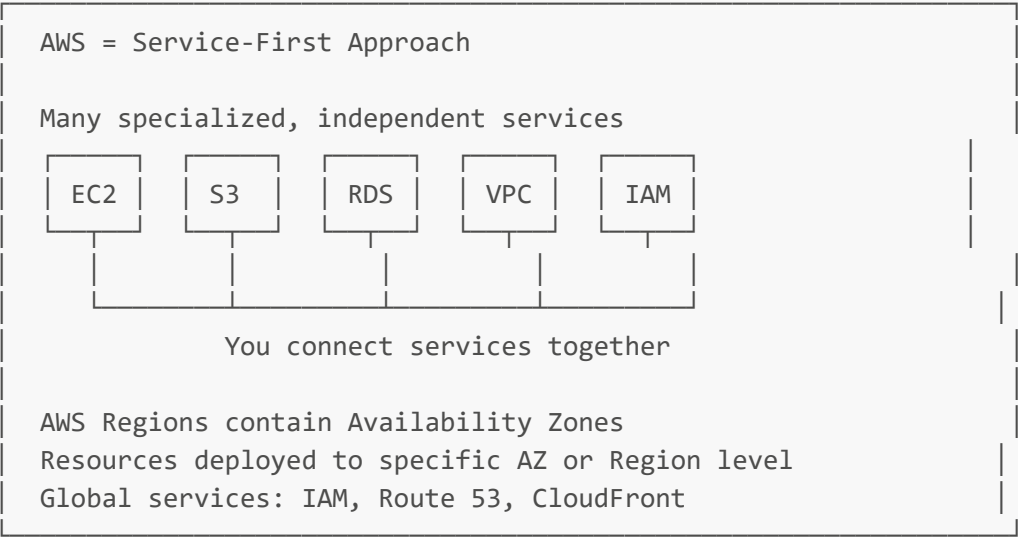
Service Name Mapping

AWS vs Azure Service Comparison		
Category	AWS	Azure
COMPUTE		
Virtual Machines	EC2	Virtual Machines
Containers	ECS / EKS	AKS / ACI
Serverless	Lambda	Azure Functions
PaaS Web Apps	Elastic Beanstalk	App Service
Batch Processing	AWS Batch	Azure Batch
STORAGE		
Object Storage	S3	Blob Storage
Block Storage	EBS	Azure Disks
File Storage	EFS	Azure Files
Archive Storage	S3 Glacier	Archive Storage
DATABASE		
Relational DB	RDS	Azure SQL
NoSQL	DynamoDB	Cosmos DB

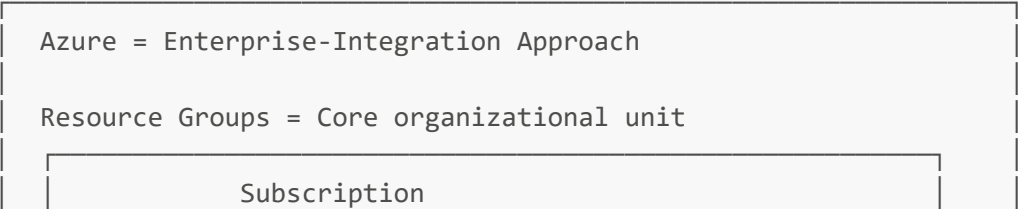
In-Memory Cache Data Warehouse	ElastiCache Redshift	Azure Cache Redis Synapse Analytics
NETWORKING Virtual Network Load Balancing DNS CDN VPN Private Connection	VPC ELB/ALB/NLB Route 53 CloudFront VPN Gateway Direct Connect	Virtual Network Azure Load Bal. Azure DNS Azure CDN VPN Gateway ExpressRoute
SECURITY & IDENTITY Identity Key Management Security Center DDoS Protection	IAM KMS Security Hub Shield	Azure AD (Entra) Key Vault Defender for Cloud Azure DDoS
MONITORING Monitoring Logging Infrastructure as Code Cost Management	CloudWatch CloudTrail CloudFormation Cost Explorer	Azure Monitor Activity Log ARM Templates / Bicep Cost Management

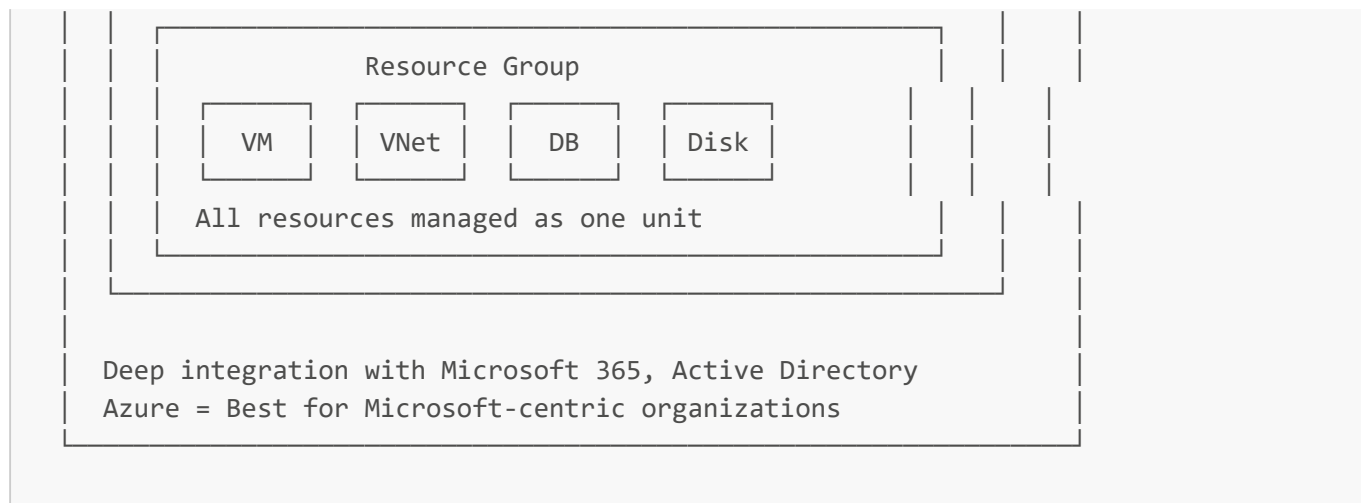
Key Architectural Differences

AWS Architecture Philosophy:



Azure Architecture Philosophy:





Pricing Model Differences

AWS Pricing:

On-Demand: Pay by the second/hour, no commitment
 Reserved Instances: 1 or 3 year commitment (up to 75% off)
 Spot Instances: Bid for unused capacity (up to 90% off)
 Savings Plans: Flexible commitment based on \$/hour

Example EC2 Pricing (t3.medium, us-east-1):

On-Demand: \$0.0416/hour = ~\$30/month
 Reserved (1yr): ~\$0.026/hour = ~\$19/month (37% savings)
 Reserved (3yr): ~\$0.016/hour = ~\$12/month (61% savings)
 Spot: ~\$0.013/hour = ~\$9/month (68% savings)

Azure Pricing:

Pay-As-You-Go: Pay by the second, no commitment
 Reserved VM Instances: 1 or 3 year (up to 72% savings)
 Spot VMs: Unused capacity (up to 90% off)
 Azure Hybrid Benefit: Use existing Windows/SQL licenses
 (Extra savings for Microsoft customers!)

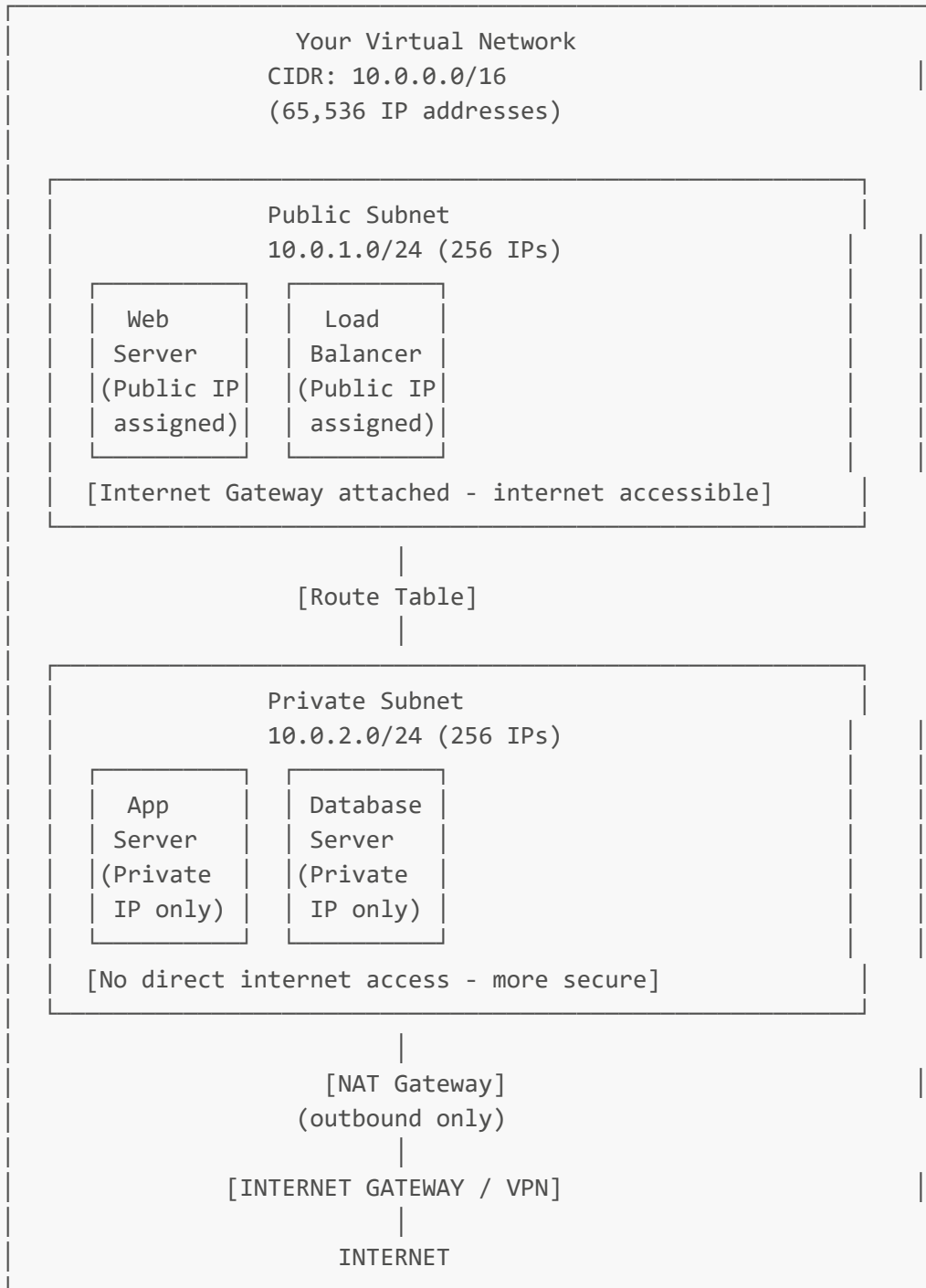
Example VM Pricing (B2s, East US):

Pay-As-You-Go: \$0.0496/hour = ~\$36/month
 Reserved (1yr): ~\$0.030/hour = ~\$22/month
 Hybrid Benefit: Additional 40% off with Windows licenses

8. Networking Fundamentals

Virtual Private Cloud/Network Concepts

VPC (AWS) / VNet (Azure) Architecture:



Security Groups vs Network Security Groups

AWS Security Groups:

Security Group = Virtual firewall for EC2 instances

Characteristics:

- ✓ Stateful (return traffic automatically allowed)
- ✓ Applied at instance level
- ✓ Allow rules only (no deny rules)

- ✓ All rules evaluated together

Example Security Group for Web Server:

Inbound Rules:

Type	Protocol	Port	Source
HTTP	TCP	80	0.0.0.0/0 (anyone)
HTTPS	TCP	443	0.0.0.0/0 (anyone)
SSH	TCP	22	203.0.113.0/32 (your IP only)

Outbound Rules:

All traffic	All	All	0.0.0.0/0 (default - allow all)
-------------	-----	-----	---------------------------------

Azure Network Security Groups (NSG):

NSG = Azure's firewall for VNets and subnets

Characteristics:

- ✓ Stateful (like AWS Security Groups)
- ✓ Applied to subnet OR network interface
- ✓ Both Allow and Deny rules
- ✓ Priority-based (lower number = higher priority)

Example NSG Rules:

Inbound Security Rules:

Priority	Name	Port	Protocol	Source	Action
100	AllowHTTP	80	TCP	*	Allow
110	AllowHTTPS	443	TCP	*	Allow
120	AllowSSH	22	TCP	My IP	Allow
65000	AllowVNet	*	*	VirtualNet	Allow
65001	AllowLB	*	*	AzureLoadB	Allow
65500	DenyAll	*	*	*	Deny

Manual Process: Create VPC in AWS

STEP 1: Navigate to VPC

1. AWS Console → Search "VPC"
2. Click "VPC" → "Your VPCs"
3. Click "Create VPC"

STEP 2: VPC Configuration

1. Resources to create:
 - VPC and more (creates subnets, route tables, gateways)
 - VPC only (manual subnet creation)

Select: • VPC and more

2. Name tag auto-generation: myapp
(Creates: myapp-vpc, myapp-subnet-public1-us-east-1a, etc.)
3. IPv4 CIDR block:
10.0.0.0/16

CIDR Notation Explained:

10.0.0.0/16 = 65,536 IP addresses available

10.0.0.0/24 = 256 IP addresses available

10.0.0.0/28 = 16 IP addresses available

Common private IP ranges:

10.0.0.0/8 (10.x.x.x)

172.16.0.0/12 (172.16.x.x - 172.31.x.x)

192.168.0.0/16 (192.168.x.x)
4. Number of Availability Zones: 2
5. Number of public subnets: 2
6. Number of private subnets: 2
7. NAT gateways: None (costs money - for learning)
In production: Select "1 per AZ" or "In 1 AZ"
8. VPC endpoints: None
9. DNS options:
 - ☒ Enable DNS hostnames
 - ☒ Enable DNS resolution

STEP 3: Review and Create

1. Review the "Preview" diagram showing:
 - VPC with CIDR
 - Public and Private subnets
 - Internet Gateway
 - Route tables
2. Click "Create VPC"
3. Wait for all resources to be created (~30 seconds)
4. Click "View VPC" when complete

9. Storage Concepts

Storage Types Explained

Three Types of Cloud Storage

1. BLOCK STORAGE (like a hard drive)

```
[ Block 1 ][ Block 2 ][ Block 3 ][Block 4]
```

OS sees it as a raw disk, can format with any filesystem
Very fast, low latency
AWS: EBS (Elastic Block Store)
Azure: Azure Managed Disks
Use case: Database storage, OS drive

2. FILE STORAGE (like a network drive)

```
/home/user/  
├─ documents/  
│   ├── report.pdf  
│   └─ data.xlsx  
└─ images/  
    └─ photo.jpg
```

Traditional file hierarchy, multiple servers can access
AWS: EFS (Elastic File System)
Azure: Azure Files
Use case: Shared content, CMS, home directories

3. OBJECT STORAGE (like a giant key-value store)

```
bucket/  
├─ images/photo1.jpg    [metadata: size, type, date]  
├─ videos/intro.mp4     [metadata: size, type, date]  
└─ backups/db-2024.zip  [metadata: size, type, date]
```

Flat structure, access via URL/API
Massive scale, cheapest, globally accessible
AWS: S3
Azure: Blob Storage
Use case: Images, videos, backups, static websites

Storage Durability & Redundancy

AWS S3 Storage Classes:

Class	Durability	Availability	Use Case
Standard	11 nines 99.999...%	99.99%	Frequent access
Standard-IA (Infrequent Access)	11 nines	99.9%	Monthly access
Intelligent-Tiering	11 nines	99.9%	Unknown

			access
Glacier Instant Retrieval	11 nines	99.9%	Quarterly access
Glacier Flexible Retrieval	11 nines	99.99% (after restore)	Archive 1-5 min retrieval
Glacier Deep Archive	11 nines	99.99% (after restore)	Long-term 12hr retrieval

10. Compute Resources

Understanding VM Instance Types

AWS EC2 Instance Family Guide:

Family	Optimized For	Example Use Cases
T	General purpose (burstable)	Dev/test, small websites, microservices
M	General purpose (balanced)	App servers, gaming servers, small databases
C	Compute optimized (more CPU)	High-performance computing, batch processing, game servers
R	Memory optimized (more RAM)	Big databases, in-memory analytics, real-time caching
G	GPU instances (graphics)	Machine learning, video encoding, 3D rendering
I	Storage optimized (fast NVMe SSD)	NoSQL databases, data warehousing
P	ML/AI optimized (NVIDIA GPUs)	Machine learning training, deep learning

Instance Size Guide:

nano < micro < small < medium < large < xlarge < 2xlarge < ...48xlarge

t3.nano: 0.5 GB RAM, 2 vCPU

t3.micro: 1 GB RAM, 2 vCPU ← Free tier

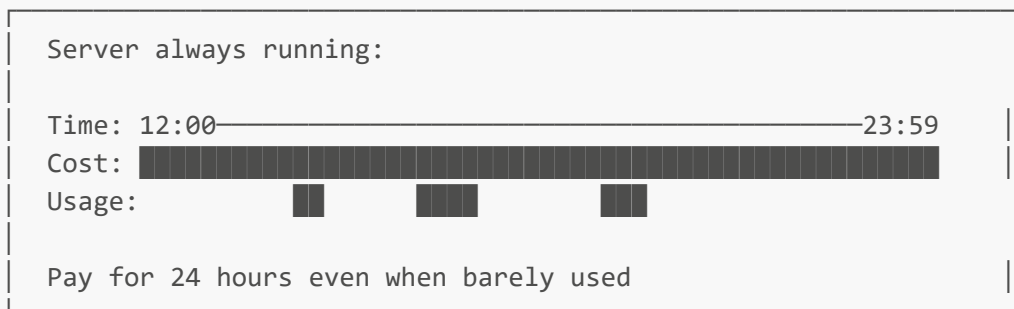
t3.small: 2 GB RAM, 2 vCPU

```
t3.medium: 4    GB RAM, 2 vCPU
t3.large:  8    GB RAM, 2 vCPU
t3.xlarge: 16   GB RAM, 4 vCPU
```

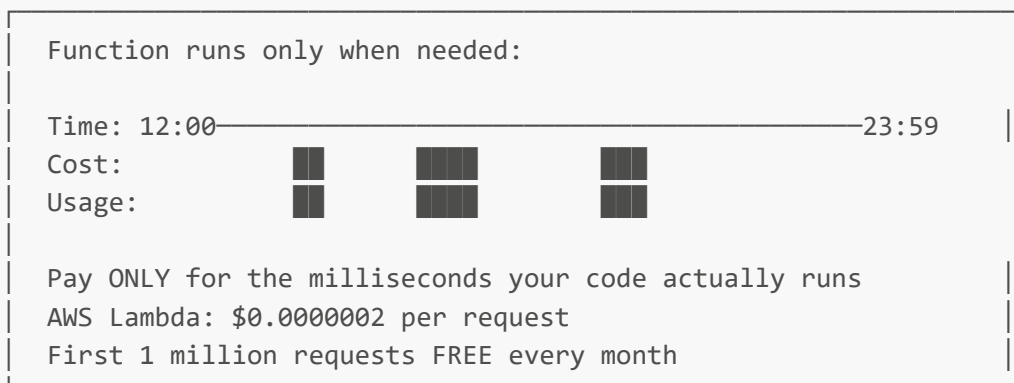
Serverless Computing Concept

Traditional (EC2/VM) vs Serverless (Lambda/Functions):

Traditional Approach:



Serverless Approach:



Serverless Use Cases:

- ✓ Image processing (resize when uploaded to S3)
- ✓ API backends (handle HTTP requests)
- ✓ Scheduled tasks (run at specific times)
- ✓ Event processing (react to database changes)
- ✓ Chatbots and automation
- ✗ Long-running processes (max 15 min for Lambda)
- ✗ Heavy computation requiring large resources
- ✗ Applications needing persistent connections

11. Security & Identity Management

AWS IAM Deep Dive

IAM Components:

IAM Structure

ROOT ACCOUNT

- └ Full access to everything - protect with MFA!

IAM USERS

- └ john@company.com (admin)
- └ jane@company.com (developer)
- └ app-service-user (application)

IAM GROUPS

- └ Administrators Group → AdministratorAccess policy
- └ Developers Group → DeveloperAccess policy
- └ ReadOnly Group → ReadOnlyAccess policy

IAM ROLES (for services and cross-account)

- └ EC2-S3-Access-Role
 - └ EC2 instances can read/write to specific S3 bucket
- └ Lambda-DynamoDB-Role
 - └ Lambda functions can access DynamoDB
- └ CrossAccount-Role
 - └ Users from Account B can access Account A

IAM POLICIES (JSON documents defining permissions)

- └ AWS Managed Policies (pre-built by Amazon)
 - └ AdministratorAccess
 - └ ReadOnlyAccess
 - └ AmazonS3FullAccess
- └ Customer Managed Policies (you create)
 - └ Custom permissions for your specific needs

Manual Process: Create IAM Policy in AWS

STEP 1: Navigate to IAM Policies

1. AWS Console → IAM → Policies (left sidebar)
2. Click "Create policy"

STEP 2: Create Policy Using Visual Editor

Method A: Visual Editor

1. Click "Add a permission"
2. Service: Search and select "S3"
3. Actions allowed:
 - Filter actions: Check boxes:
 - ☒ ListBucket (see what's in bucket)
 - ☒ GetObject (download files)

☒ PutObject (upload files)

Leave others unchecked

4. Resources: Choose specific bucket

- bucket: Add ARN

Bucket name: my-specific-bucket

- object: Add ARN

Bucket name: my-specific-bucket

Object name: * (all objects)

5. Click "Next"

Method B: JSON Editor (copy-paste this example)

Click "JSON" tab:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "s3:ListBucket"
      ],
      "Resource": "arn:aws:s3:::my-specific-bucket"
    },
    {
      "Effect": "Allow",
      "Action": [
        "s3:GetObject",
        "s3:PutObject",
        "s3:DeleteObject"
      ],
      "Resource": "arn:aws:s3:::my-specific-bucket/*"
    }
  ]
}
```

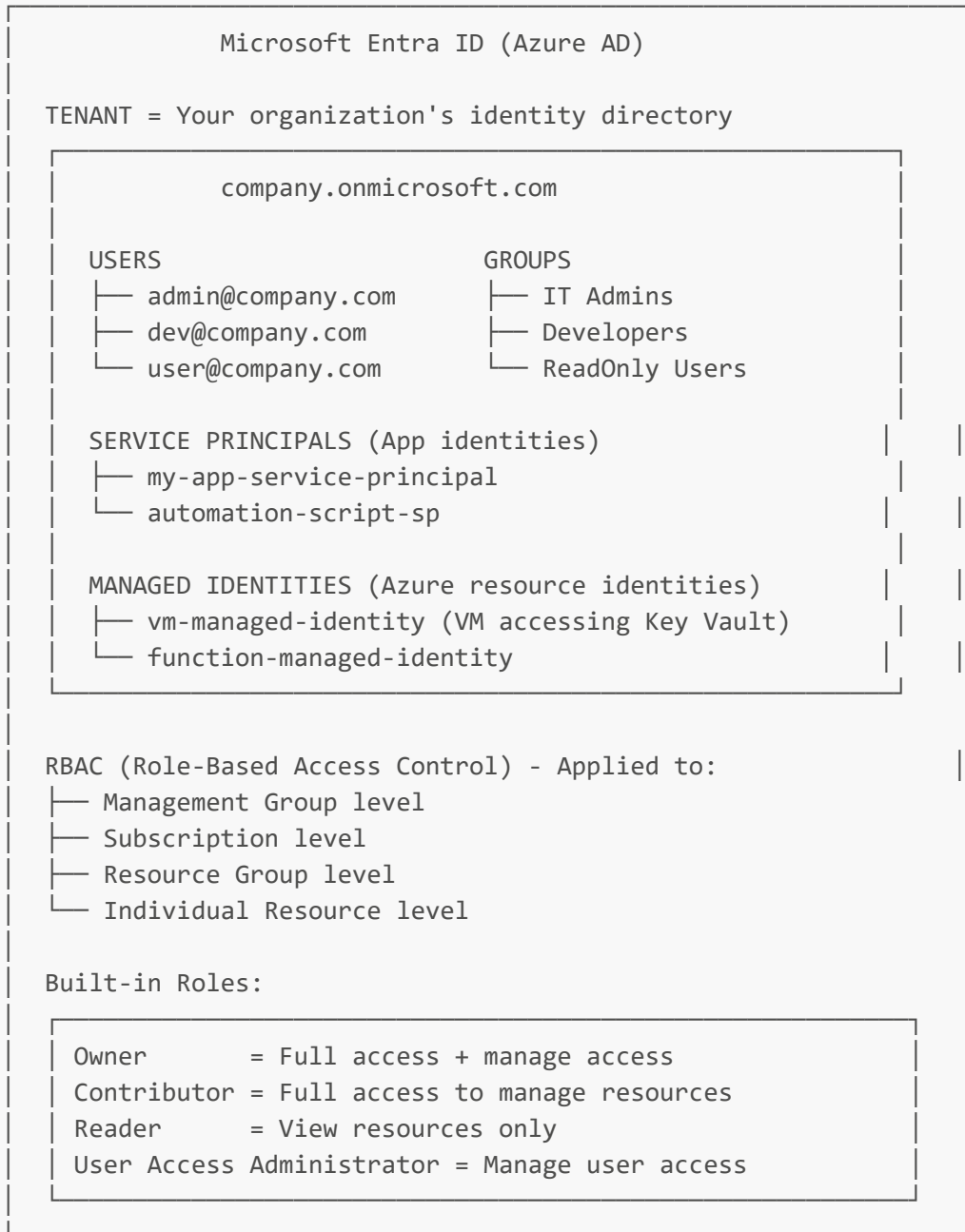
STEP 3: Policy Details

1. Policy name: S3-MyBucket-ReadWrite
2. Description: Allow read and write access to my-specific-bucket
3. Click "Create policy"

STEP 4: Attach Policy to User/Group/Role

1. IAM → Users → Select user
2. Click "Add permissions"
3. Click "Attach policies directly"
4. Search: S3-MyBucket-ReadWrite
5. ☒ Check the policy
6. Click "Next" → "Add permissions"

Azure Identity Structure:



Manual Process: Assign RBAC Role in Azure

STEP 1: Navigate to Resource Group

1. Azure Portal → Resource groups
2. Click on your resource group: rg-myapp-dev

STEP 2: Access Control (IAM)

1. In left sidebar, click "Access control (IAM)"
2. You see current role assignments
3. Click "+ Add" → "Add role assignment"

STEP 3: Select Role

1. "Role" tab shows:

```
Job function roles:
  Owner
  Contributor    ← Good for developers
  Reader         ← Good for auditors
  ...and hundreds more

Privileged administrator roles:
  Role Based Access Control Administrator
  User Access Administrator
```

2. Select "Contributor"

3. Click "Next"

STEP 4: Select Member (Who Gets the Role)

1. Assign access to:

- User, group, or service principal

2. Click "+ Select members"

3. Search for user by name or email

4. Click on user to select them

5. Click "Select"

6. Click "Next"

STEP 5: Review + Assign

1. Review:

- Role: Contributor
- Scope: /subscriptions/.../resourceGroups/rg-myapp-dev
- Member: user@company.com

2. Click "Review + assign"

3. Click "Review + assign" again to confirm

4. Role assigned successfully

12. Monitoring & Management

AWS CloudWatch

CloudWatch Overview:

AWS CloudWatch

METRICS - Numerical data over time

EC2 Metrics (collected every 5 min free, 1 min paid):

- CPUUtilization (0-100%)
- NetworkIn / NetworkOut (bytes)

- DiskReadOps / DiskWriteOps
- StatusCheckFailed

Custom Metrics: You can push YOUR app metrics

ALARMS - Alert when metric crosses threshold

Example Alarm:

Metric: CPUUtilization

Threshold: > 80% for 5 consecutive minutes

Action: Send email notification

OR Auto-scale (add more instances)

LOGS - Application and system logs

Log Groups → Log Streams → Log Events

/aws/lambda/my-function

└─ 2024/01/15/[LATEST]abc123...

└─ 2024-01-15 10:30:15 Function started

└─ 2024-01-15 10:30:16 Processing data...

└─ 2024-01-15 10:30:17 Function completed

Manual Process: Create CloudWatch Alarm (AWS)

STEP 1: Navigate to CloudWatch

1. AWS Console → Search "CloudWatch"
2. Click "CloudWatch"
3. Left sidebar → "Alarms" → "All alarms"
4. Click "Create alarm"

STEP 2: Select Metric

1. Click "Select metric"
2. Browse or search metrics:
 - Click "EC2" → "Per-Instance Metrics"
 - Find your instance ID
 - Click "CPUUtilization"
3. Click "Select metric"

STEP 3: Specify Metric and Conditions

1. Metric name: CPUUtilization
2. Statistic: Average
3. Period: 5 minutes

4. Conditions:

- Threshold type: • Static
- Whenever CPUUtilization is:
 - Greater than
- than: 80

This means: Alert when CPU averages > 80%
for one 5-minute period

5. Click "Next"

STEP 4: Configure Actions

1. Alarm state trigger: • In alarm
2. Send notification to:
 - Select an SNS topic OR
 - "Create new topic"
 - Topic name: high-cpu-alerts
 - Email endpoints: your@email.com
 - Click "Create topic"
3. Click "Next"

STEP 5: Name and Description

1. Alarm name: High-CPU-Warning-MyServer
2. Alarm description: Alert when EC2 CPU > 80%
3. Click "Next" → "Create alarm"

STEP 6: Confirm Email Subscription

1. Check your email
2. Email from AWS: "AWS Notification - Subscription Confirmation"
3. Click "Confirm subscription" link
4. ☒ You'll now receive email when CPU exceeds 80%

Manual Process: Create Alert in Azure Monitor

STEP 1: Navigate to Azure Monitor

1. Azure Portal → Search "Monitor"
2. Click "Monitor"
3. Left sidebar → "Alerts"
4. Click "+ Create" → "Alert rule"

STEP 2: Select Scope (Resource to Monitor)

1. Click "+ Select scope"
2. Filter by:
 - Resource type: Virtual machines
3. Select your VM from the list

4. Click "Done"

STEP 3: Configure Condition (What to Monitor)

1. Click "+ Add condition"
2. Search signals: "Percentage CPU"
3. Click "Percentage CPU"
4. Configure signal logic:
 - Threshold: Static
 - Aggregation type: Average
 - Operator: Greater than
 - Threshold value: 80
 - Check every: 5 minutes
 - Lookback period: 5 minutes
5. Click "Add"

STEP 4: Actions (What to Do When Alert Fires)

1. Click "+ Create action group"
2. Action group basics:
 - Subscription: Your subscription
 - Resource group: rg-myapp-dev
 - Action group name: email-alerts-group
 - Display name: EmailAlerts
3. Notifications:
 - Notification type: Email/SMS message/Push/Voice
 - Name: Email-Notification
 - Click edit (pencil icon):
 - ☒ Email: your@email.com
 - Click "OK"
4. Click "Review + create" → "Create"

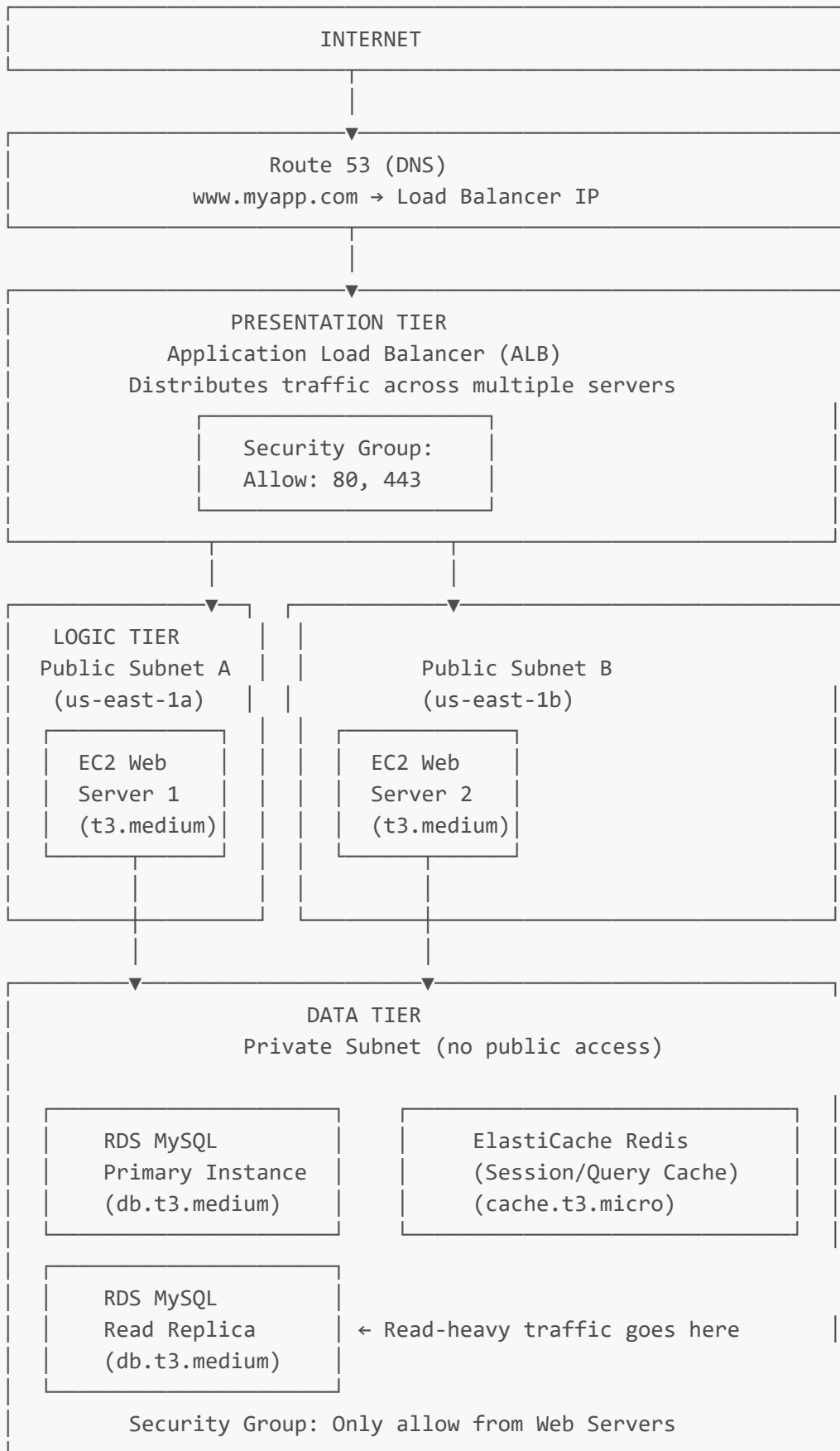
STEP 5: Alert Rule Details

1. Alert rule name: High-CPU-Alert-MyVM
2. Description: Alert when VM CPU > 80%
3. Severity:
 - 0 = Critical
 - 1 = Error
 - 2 = Warning ← Select this
 - 3 = Informational
4. Region: East US
5. Click "Review + create" → "Create"

13. Real-World Architecture Patterns

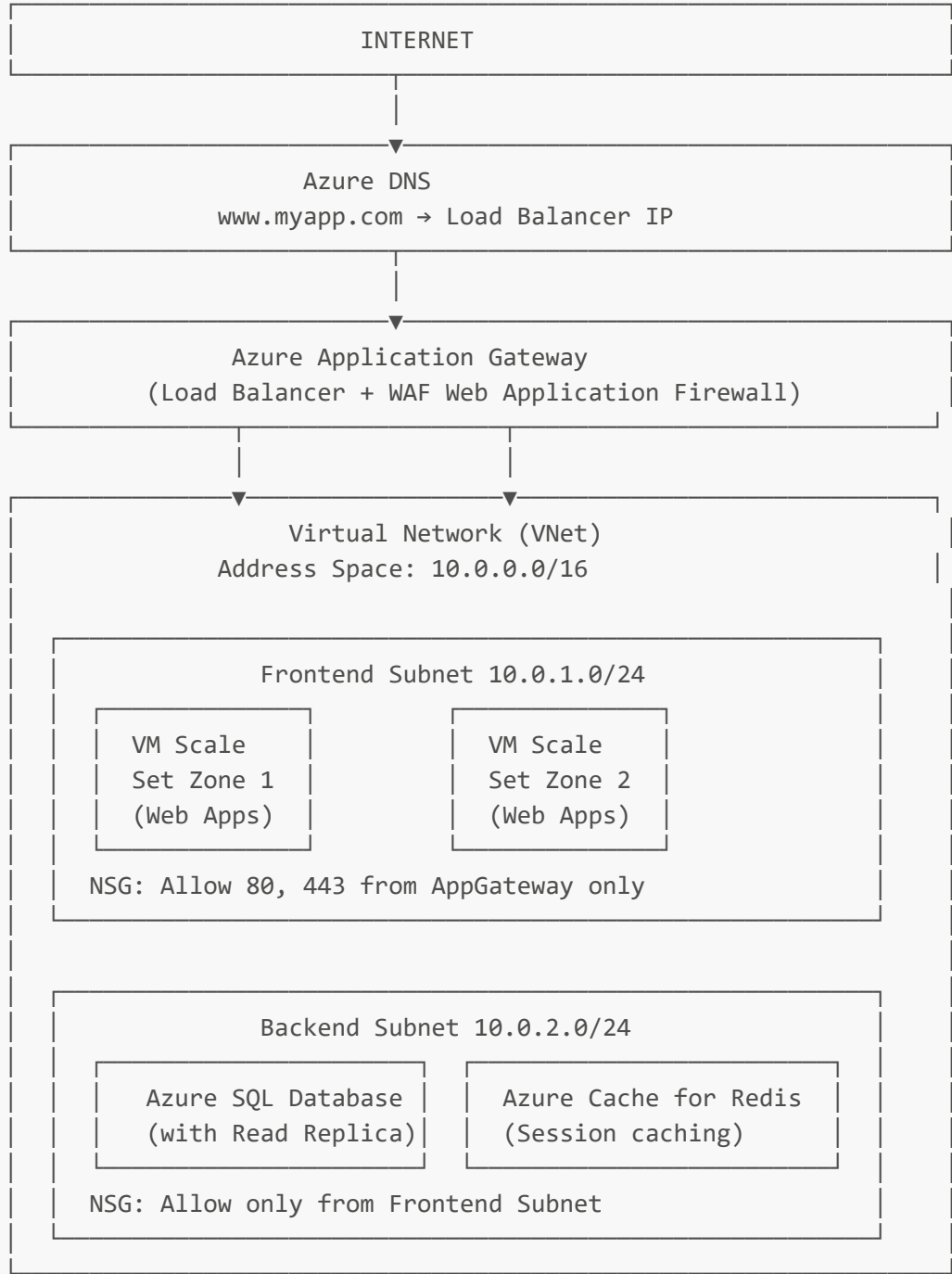
Three-Tier Web Application Architecture

AWS Three-Tier Architecture:



Azure Equivalent Architecture

Azure Three-Tier Architecture:



Cost Estimation Guide

Monthly Cost Estimation - Small Web Application:

AWS Estimate:

Service	Specs	Monthly Cost
EC2 (2 instances)	t3.medium	\$60 (\$30 each)
Application LB	Standard	\$16
RDS MySQL	db.t3.small	\$26

S3 Storage	50 GB	\$1
Data Transfer	10 GB out	\$0.90
Route 53	1 hosted zone	\$0.50
CloudWatch	Basic monitoring	\$0
TOTAL ESTIMATE		~\$104/month

Azure Estimate:

Service	Specs	Monthly Cost
VMs (2 instances)	B2s	\$72 (\$36 each)
Application Gateway	Standard v2 WAF	\$117
Azure SQL Database	S1	\$15
Blob Storage	50 GB LRS	\$1
Data Transfer	10 GB out	\$0.87
Azure DNS	1 zone	\$0.90
Azure Monitor	Basic	\$0
TOTAL ESTIMATE	(App Gateway expensive tier)	~\$207/month

⚠ Use official calculators for accurate pricing:
AWS: <https://calculator.aws>
Azure: <https://azure.microsoft.com/pricing/calculator>

Summary: Key Concepts Cheat Sheet

CLOUD COMPUTING CHEAT SHEET
SERVICE MODELS IaaS = YOU manage OS, middleware, apps, data PaaS = YOU manage apps and data only SaaS = YOU manage nothing (just use it)
DEPLOYMENT MODELS Public = Shared infrastructure (AWS, Azure, GCP) Private = Dedicated infrastructure (your org only) Hybrid = Both public and private connected Multi = Multiple cloud providers
AWS KEY CONCEPTS Region = Geographic area (us-east-1) AZ = Availability Zone = Data center within region VPC = Virtual Private Cloud = Your private network

EC2 = Virtual machines

S3 = Object storage

IAM = Identity and Access Management

AZURE KEY CONCEPTS

Region = Geographic area (East US)

Availability Zone = Data centers within region

VNet = Virtual Network = Your private network

Virtual Machines = Compute instances

Blob Storage = Object storage

Resource Group = Container for related resources

Azure AD/Entra = Identity management

RBAC = Role-Based Access Control

STORAGE TYPES

Block = Raw storage for databases (EBS/Azure Disks)

File = Shared file system (EFS/Azure Files)

Object = Unlimited file storage (S3/Blob Storage)

SECURITY BEST PRACTICES

- ✓ Enable MFA on all accounts
- ✓ Use IAM roles instead of access keys
- ✓ Follow principle of least privilege
- ✓ Keep sensitive data in private subnets
- ✓ Encrypt data at rest and in transit
- ✓ Enable logging and monitoring
- ✓ Use security groups to restrict access

Learning Resources & Next Steps

Recommended Learning Path

BEGINNER (Months 1-2):

- Complete this guide and set up accounts
- Launch EC2 instance and S3 bucket
- Create Azure VM and Storage Account
- Understand VPC/VNet networking basics
- Learn IAM/RBAC fundamentals

INTERMEDIATE (Months 3-4):

- Deploy 3-tier application on AWS or Azure
- Set up auto-scaling and load balancing
- Implement monitoring and alerting
- Learn Infrastructure as Code (CloudFormation/Terraform)
- Study AWS Solutions Architect Associate or Azure Administrator Associate certification

ADVANCED (Months 5-6):

- ❑ Container orchestration (Kubernetes/EKS/AKS)
- ❑ Serverless architectures (Lambda/Azure Functions)
- ❑ CI/CD pipelines (CodePipeline/Azure DevOps)
- ❑ Multi-region disaster recovery
- ❑ Cost optimization strategies

FREE PRACTICE RESOURCES:

AWS Free Tier: <https://aws.amazon.com/free>

Azure Free: <https://azure.microsoft.com/free>

AWS Training: <https://aws.amazon.com/training/digital>

Microsoft Learn: <https://learn.microsoft.com>

A Cloud Guru: <https://acloudguru.com> (paid)

This guide covers foundational cloud computing concepts with manual implementation steps for both AWS and Azure. Practice each section in the free tiers before moving to production workloads. Always delete resources you're no longer using to avoid unexpected charges.